Nonlinear Solvers	
Overview	
• compute the numerical inverse of a function using Newton's method • **compute partial derivatives and compute the first three terms of a Taylor series approximation for a function of multiple arguments** • identify the three terms for the Taylor series in terms of an approximation of a function multiple arguments • **relate Taylor series terms to tangent plane and critical points** • applying Newtons method in higher dimensions to solve a nonlinear system of equations • **the convergence challenges for Newtons method** • why Newton's method is not the complete solution for a nonlinear solver • **the data structures used for forward differentiation** • use numerical libraries that implement the forward differentiation approach	
Async	
Nonlinear Solvers in 1-D Part 2	
Use Case	• **p**robability **d**ensity **f**unction for the **normal distribution** for a given center and **standard deviation** $$f(x) = \frac{1}{\sigma\sqrt{(2\pi)}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$ Probability Density Function • **probability** that a value is **less** than x Cumulative Distribution Function $$\Phi(x) = \frac{1}{\sigma\sqrt{(2\pi)}} \int_{-\infty}^{x} e^{-\frac{1}{2}\left(\frac{t-\mu}{\sigma}\right)^2} dt = \frac{1}{2}\left[1 + \text{erf}\left(\frac{x-\mu}{\sigma\sqrt{(2)}}\right)\right]$$ •

	• where $erf(x)$ is a special **function defined** by erf = error funciton Error Function $$\mathbf{erf}(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}dt$$
Invert	• to find the **inverse** we view it as a **root finding problem** $$y = \mathrm{erf}(x) \rightarrow f(x) = \mathrm{erf}(x) - y = 0$$ • we need an **initial guess**, and for most x $\mathrm{erf}(x)$ is close to x, so $$x_0 = y$$ • for **Newtons method** we need the **derivative** of $\mathrm{erf}(x)$, but that comes from the **fundamental theorem** of **calculus** $$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}dt \rightarrow erf'(x) = \frac{2}{\sqrt{\pi}}e^{-x^2}$$ • which leads to the **iteration** $$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{\mathrm{erf}(x_n) - y}{\frac{2}{\sqrt{\pi}}e^{-x_n^2}}$$ $$= x_n - \frac{\sqrt{\pi}}{2}e^{x_n^2}(\mathrm{erf}(x_n) - y)$$

	`julia> using SpecialFunctions`
	https://specialfunctions.juliamath.org/v0.1/
	`julia> y = 0.5; x = y;`
	`julia> x = x - \sqrt(\pi)/2 * exp(x^2) * (erf(x) - y)`
	`0.47667241214786027`
	`julia> x = x - \sqrt(\pi)/2 * exp(x^2) * (erf(x) - y)`
	`0.47693624301317533`

	`julia> x = x - \sqrt(\pi)/2 * exp(x^2) * (erf(x) - y)`
	`0.4769362762044694`
	
	`julia> x = x - \sqrt(\pi)/2 * exp(x^2) * (erf(x) - y)`
	`0.4769362762044699`
	
	`julia> x = x - \sqrt(\pi)/2 * exp(x^2) * (erf(x) - y)`
	`0.4769362762044699`
	
	`julia> erf(x)`
	`0.5`

Taylor Series in Higher Dimensions	
Taylor Series	• function of **one argument** the Taylor Series $$f(x) = f(a) + f'(a)(x-a) + \cdots + \frac{f^{((n))}(a)}{n!}(x-a)^n + \frac{f^{((n+1))}(a)}{(n+1)!}(\xi - a)^{n+1}$$ • consider the case of a **vector valued function** of one variable $$f : R \rightarrow R^n$$ • component by component, **vector valued derivative** $$f : R \rightarrow R^m$$ $$f(t) = [t^2;\ t + \sin(t)]$$ $$f'(t) = [2t;\ 1 + \cos(t)]$$ $$f(t) = [a^2;\ a + \sin(a)] + [2a;\ 1 + \cos(a)](t-a) + \frac{[2;\ -\sin(a)]((t-a)^2)}{2!} + \cdots$$

- **Taylor Series in 2D**

- now we allow the **scalar** function to have **multiple** input **arguments**

$$F(x,y) = F\big(x_0 + (x - x_0), y_0 + (y - y_0)\big)$$

- **focus** on the **first** argument, and keep the **second** argument **fixed**

$$F(x,y) = F\big(x_0 + (x - x_0), y\big)$$

  - now that we have a **function** of a **single argument**, apply the standard **Taylor's Series**

as a **function** of y:

represents the tangent plane approximation of the function at the point (x_0, y_0) along the $y - axis$

$$F(x_0,y) = \underbrace{F\big(x_0, y_0 + (y - y_0)\big)}_{f(y_0 + (y - y_0))} = \overbrace{\underbrace{F(x_0,y_0)}_{f(y_0)}}^{\text{constant}} + \overbrace{\underbrace{\frac{\partial F}{\partial y}(x_0,y_0)(y - y_0)}_{f'(y_0)(y - y_0)}}^{\text{linear term}} + \overbrace{\underbrace{\frac{\partial^2 F}{\partial y^2}(x_0,y_0)\frac{((y - y_0)^2)}{2!}}_{\frac{f''(y_0)((y - y_0)^2)}{2!}}}^{\text{quadratic}} + \cdots$$

Partial Differentiation:
by holding x constant, you treat the multivariable function $F(x_0, y)$ as a univariate function $f(y)$

this allows the use of standard Taylor expansion rules using partial derivatives $\frac{\partial F}{\partial y}$

by symmetry, the expansion along the $x - axis$ is identical, substituting all y terms for x and holding y constant

a little busy

The Symmetry Rule

- the **Taylor expansion** for any **single variable** is performed by **holding** all **other** variables **constant** at the **anchor** point (x_0, y_0)

Axis of Expansion	Variable Held Constant	Resulting Univariate Function	First-Order (Linear) Term
y-axis	$x = x_0$	$f(y) = F(x_0, y)$	$\frac{\partial F}{\partial y}(x_0, y_0)(y - y_0)$
x-axis	$y = y_0$	$f(x) = F(x, y_0)$	$\frac{\partial F}{\partial x}(x_0, y_0)(x - x_0)$

Combine

- group by powers, and skip (x_0, y_0)

1. the **single variables** start (along x)

$$F(x,y) = F(x_0,y) + \frac{\partial F}{\partial x}(x_0,y)(x - x_0) + \frac{\partial^2 F}{\partial x^2}(x_0,y)\frac{((x - x_0)^2)}{2!} + \cdots$$

2. the **full** bivariate **expansion** (**all** terms)

$$F(x,y) = F(x_0,y_0) + \frac{\partial F}{\partial x}(x - x_0) + \frac{\partial F}{\partial y}(y - y_0)$$

$$+\frac{1}{2}\left[\frac{\partial^2 F}{(\partial x^2)(x - x_0)^2} + \frac{2(\partial^2 F)}{(\partial x\, \partial y)(x - x_0)(y - y_0)} + \frac{\partial^2 F}{(\partial y^2)(y - y_0)^2}\right]$$

3. the **matrix** wrap-up (**quadratic form**)

$$F(x,y) = F(x_0,y_0) + \left[\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}\right][x - x_0;\ y - y_0]$$

$$+\frac{1}{2}[x - x_0, y - y_0]\left[\frac{\partial^2 F}{\partial x^2}, \frac{\partial^2 F}{\partial x\, \partial y}; \frac{\partial^2 F}{\partial x\, \partial y}, \frac{\partial^2 F}{\partial y^2}\right][x - x_0;\ y - y_0]$$

Approximations

Tangent Plane

- Taylor Series in two dimensions for a scalar valued function is

$$f(x_0, y_0) + \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right] \begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix}$$

$$+ \frac{1}{2}\begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix}^T \begin{bmatrix} \frac{\partial^2 f}{\partial x^2}, & \frac{\partial^2 f}{\partial x\,\partial y} \\ \frac{\partial^2 f}{\partial x\,\partial y}, & \frac{\partial^2 f}{\partial y^2} \end{bmatrix} \begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix} + \cdots$$

- approximations handles what happens close by
    - the point the second term shrinks faster than the first

$$f(x, y) = f(x_0, y_0) + \left(\frac{\partial f}{\partial x}(x_0, y_0), \frac{\partial f}{\partial y}(x_0, y_0)\right) \cdot (x - x_0, y - y_0)$$

$$z = a + (b, c) \cdot (x - x_0, y - y_0)$$

plane equation → $-bx - cy + 1z = a - bx_0 - cy_0 = C$

General

- this is **not limited** to **two dimensions**

nor does it require the **number** of **input arguments** to be the **same** as output

$$F\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} \cos(x) + y \\ x^2 + 2y^2 - z \end{bmatrix}$$

$$F\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) \approx F\left(\begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix}\right) + \underbrace{\begin{bmatrix} -\sin(x) & 1 & 0 \\ 2x & 4y & -1 \end{bmatrix}}_{\text{Jacobian}} \begin{bmatrix} x - x_0 \\ y - y_0 \\ z - z_0 \end{bmatrix}$$